Evaluating Temporal Link Prediction methods to generate synthetic social networks data

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February 18, 2026

Problem Statement 🎯

- Social networks critical for understanding modern online communities
- Risky users exploit platforms for malicious activities
- Temporal dimension crucial for capturing evolving patterns
- Labeled data scarcity hinders supervised learning approaches
- **Goal**: Temporal link prediction for social network analysis on synthetic data

Objectives

1. Development of a temporal link prediction model
2. Quantitative comparison of model performance
3. Analysis of network evolution patterns and structural properties
4. Application of the trained model to generate synthetic network links

Dataset: Gab Social Network 📊

- Platform: Gab (2016-2025)
- 6 temporal snapshots
- Posts + social network graphs
- 768-dimensional BERT embeddings for every post

Property	Value
Total Posts	182,545
Unique Users	27,002
Snapshots	6
Time Span	2016-2025

Synthetic Dataset 🤖

Synthetic Users

- 3 batches: $400 + 400 + 200$
- Total: 1,000 synthetic users
- 9,990 synthetic posts
- 768-dimensional BERT embeddings for every post

Purpose

- Test model generalization
- Generate synthetic network links
- Evaluate structural realism

Data Quality 🔍

Attribute	Missing %
Language	0.48%
All others	0.00%

- Posts lacking textual content were removed.
- Posts containing images or videos were removed.
- Posts written in languages other than English were removed.
- HTML tags were stripped from the remaining content.

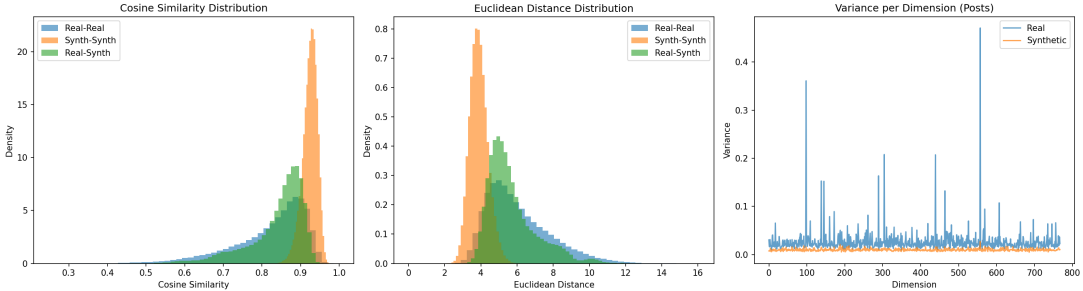
Snapshot	Snap.	Incr.	Cov.
2016-2021	18,926	18,926	100%
2022	13,108	32,034	100%
2023	13,376	45,410	100%
2024	13,400	58,810	100%
Jan-Jul 25	44,878	103,688	100%
Jul 25	78,857	182,545	100%

Duplicate Content Analysis 🔍

Snapshot	Dup. %
2016-2021	1.79%
2022	3.02%
2023	4.21%
2024	4.20%
Jan-Jul 25	7.52%
Jul 25	10.04%

- Increasing duplicate rate over time
- No duplicate post IDs
- Each post has unique metadata
- Reblogs as standalone posts

BERT Embeddings: Real vs Synthetic 📈



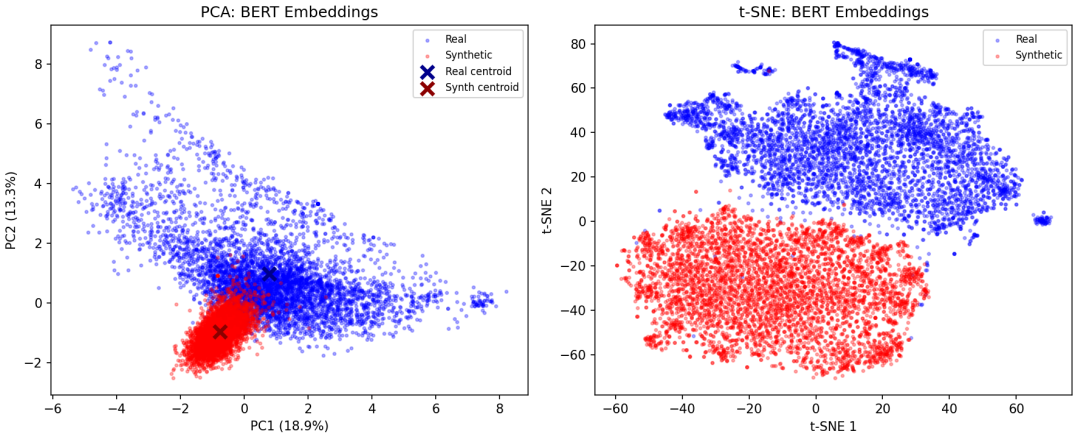
Embedding Statistics 🧮

Metric	Real	Synthetic
Posts	182,545	9,990
Cos. Sim. (mean)	0.821	0.924
Cos. Sim. (std)	0.097	0.019
Variance	0.026	0.010

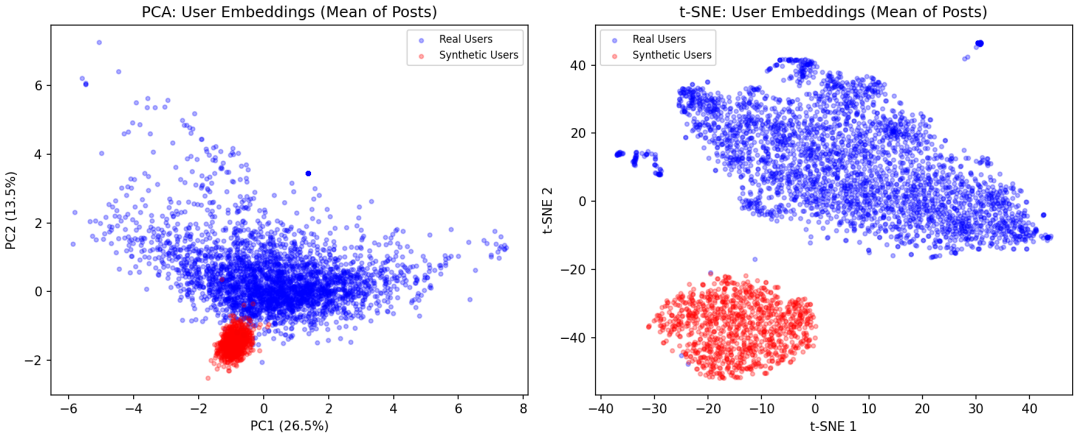
Key Finding

Synthetic users: variance ratio 11.23× lower than real users

Embedding Space Visualization



User-Level Embedding Analysis 👥



Feature Construction

Node features from BERT embeddings:

$$\mathbf{h}_u^{(t)} = \frac{1}{|P_u^{(t)}|} \sum_{p \in P_u^{(t)}} \text{BERT}(\text{content}_p)$$

Graph Structure

- **Directed**, **unweighted** graph $G_t = (V_t, E_t)$
- Cumulative edge assumption
- Sparse adjacency representation

EvolveGCN Architecture

Graph Convolution:

$$H_t^{(l+1)} = \sigma(\hat{A}_t H_t^{(l)} W_t^{(l)})$$

Weight Evolution (EvolveGCN-H):

$$W_t^{(l)} = \mathrm{GRU}(H_t^{(l)}, W_{t-1}^{(l)})$$

- Handles new nodes at test time
- No requirement for static node sets

Key Innovation

RNN evolves GCN parameters instead of node embeddings

Top-K Selection

$$\text{scores} = \frac{H_t^{(l)} \cdot s}{\|s\|}$$

Select most relevant embeddings for weight evolution

GCN Architecture Details

GRCU Layers:

- Layer 1: 768 $\rightarrow$ 128
- Layer 2: 128 $\rightarrow$ 64

MLP Classifier:

- Input: 128 (64 $\times$ 2 concatenated)
- Output: 2 logits

Loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_1 y_i \log(\hat{y}_i) + w_0(1-y_i)\log(1-\hat{y}_i)\right]$$

Training Ingredients 🛒

Training Input:

- Historical adjacency list $[\hat{A}_{t-k}, \ldots, \hat{A}_{t-1}]$
- Historical node features $[H_{t-k}, \ldots, H_{t-1}]$
- Node masks for active nodes
- Negative sampling considering only active nodes and history

Training Output:

- Edge pairs (u, v)
- Binary labels $y \in \{0, 1\}$

Training Pipelines 🔁

Original Implementation

- Sequence-based training
- All snapshots in single batch
- 50 epochs total
- GRU manages temporal flow

Incremental Implementation

- Phase-based training
- One snapshot per phase
- 50 epochs per phase
- Weight transfer between phases

Configurations: Learning Rate $\in \{0.001, 0.0005\}$, Negative Multiplier $\in \{1, 2\}$

Classification Metrics

MAP (Mean Average Precision)

$$\mathrm{MAP} = \frac{1}{Q} \sum_{q=1}^{Q} \mathrm{AP}(q)$$

Mean of per-query average precisions; ranking quality metric.

AUC-ROC

$$\mathrm{AUC} = \frac{1}{N_1 N_0} \sum_{i=1}^{N_1} \sum_{j=1}^{N_0} \mathbb{I}(p_i > p_j)$$

Prob. that a positive ranks higher than a negative.

Macro Metrics

Precision:

$$\frac{1}{C} \sum_{c=1}^{C} \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}$$

Recall:

$$\frac{1}{C} \sum_{c=1}^{C} \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}$$

F1:

$$\frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

Graph Structure Metrics

Average Degree

$$\text{AvgDeg} = \frac{|E|}{|V|}$$

Mean connections per node; density indicator.

Clustering Coefficient

$$C_i = \frac{2e_i}{k_i(k_i - 1)}$$

Local transitivity; triangle closure tendency.

Average Path Length

$$\text{AvgSP} = \frac{1}{|P|} \sum_{(i,j)} d(i,j)$$

Typical node distance; testing the small-world property.

Modularity

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

Community structure quality (-1 to 1).

Classification Results 📊

	LR	NM	MAP	AUC	Mac P	Mac R	Mac F1
Incr.	0.001	1	0.78	0.78	0.25	0.49	0.33
	0.001	2	0.64	0.74	0.75	0.68	0.69
	0.0005	1	0.78	**0.80**	0.68	0.52	0.38
	0.0005	2	0.62	0.72	0.73	0.68	0.69
Orig.	0.001	1	**0.87**	0.63	**0.86**	**0.86**	**0.86**
	0.001	2	0.77	0.65	0.82	0.83	0.83
	0.0005	1	0.86	0.64	0.82	0.79	0.78
	0.0005	2	0.76	0.65	0.81	0.84	0.81

*Incr. = **Incremental**, Orig. = **Original**, LR = Learning Rate, NM = Negative Multiplier*

Per-Class Metrics 📊

	LR	NM	Class 0 (Non-link)			Class 1 (Link)		
			P	R	F1	P	R	F1
Incr.	0.001	1	0.00	0.00	0.00	0.49	0.97	0.65
	0.001	2	0.77	**0.92**	0.83	0.73	0.44	0.55
	0.0005	1	0.85	0.04	0.08	0.51	**0.99**	0.67
	0.0005	2	0.77	0.90	0.83	0.69	0.47	0.56
Orig.	0.001	1	0.89	0.83	0.86	**0.84**	0.89	**0.87**
	0.001	2	0.89	0.88	**0.88**	0.76	0.78	0.77
	0.0005	1	0.93	0.62	0.74	0.71	0.96	0.82
	0.0005	2	**0.95**	0.76	0.84	0.66	0.93	0.77

Incr. = ***Incremental***, *Orig.* = ***Original***, *LR* = *Learning Rate*, *NM* = *Negative Multiplier*

Graph Metrics Comparison

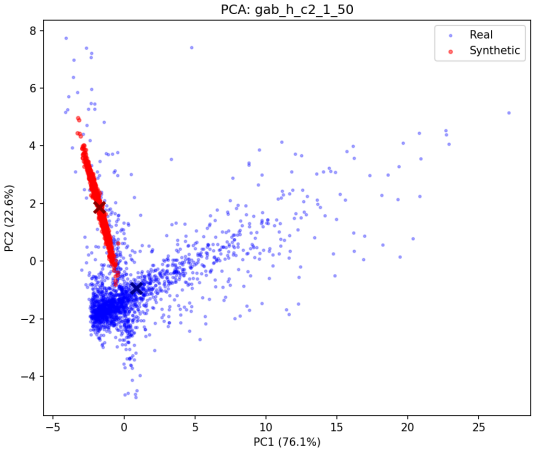
Network	Avg Deg	Mod.	Avg Path	Clust.
Facebook	43.69	0.83	3.62	0.61
Twitter	21.75	0.80	4.90	0.57
Google+	127.06	0.47	3.32	0.49
Gab (Ours)	32.11	0.27	3.04	0.08
Incr., LR=0.0005, NM=1	220.35	-0.001	1.02	0.59

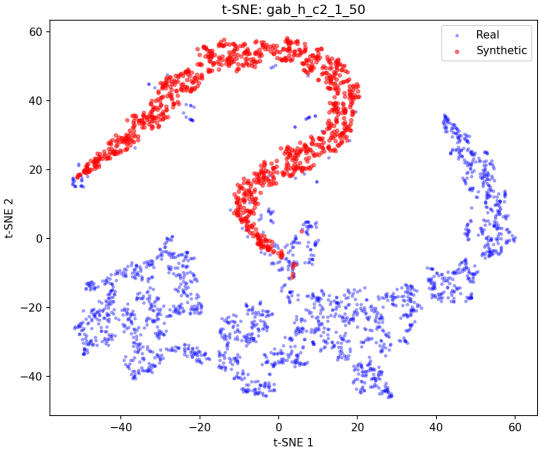
Avg Deg = Average Degree, Mod. = Modularity, Avg Path = Average Path Length, Clust. = Clustering Coefficient

PCA Statistics 📊

	Config	PC1	PC2	Total	Centroid
Orig.	LR=0.001, NM=1	99.98%	0.01%	100.00%	6.29
	LR=0.001, NM=2	99.71%	0.23%	99.94%	2.33
	LR=0.0005, NM=1	95.99%	3.39%	99.38%	2.06
	LR=0.0005, NM=2	99.96%	0.03%	100.00%	6.65
Incr.	LR=0.001, NM=1	99.65%	0.21%	99.86%	1.01
	LR=0.001, NM=2	99.39%	0.57%	99.96%	3.41
	LR=0.0005, NM=1	76.13%	22.61%	98.74%	3.84
	LR=0.0005, NM=2	91.20%	8.47%	99.67%	4.78

PCA Statistics





Overall Assessment

Achieved

- Temporal link prediction model
- Quantitative evaluation

Failed

- Realistic synthetic networks
- Generalization to synthetic users

Future Work

1. **Synthetic Data Improvement**
   - Generate synthetic posts with more diverse prompts
   - Apply embedding perturbation techniques to increase variance
   - Reduce isolation of synthetic users in embedding space
2. **Test Other Models**
   - Explore alternative temporal graph models
3. **Explore New Metrics**
   - Custom loss functions for synthetic user link prediction
   - Penalize embedding isolation
   - Encourage diversity in synthetic representations

Thank You 🚀